



FOOD SCIENCE AND TECHNOLOGY SAMPLE EXAMINATION STAGE 2

Section 7 of the *New WACE Manual: General Information 2006–2009* outlines the policy on WACE examinations.

Further information about the WACE Examinations policy can be accessed from the Curriculum Council website at http://newwace.curriculum.wa.edu.au/pages/about_wace_manual.asp.

The purpose for providing a sample examination is to provide teachers with an example of how the course will be examined. Further finetuning will be made to this sample in 2008 by the examination panel following consultation with teachers, measurement specialists and advice from the Assessment, Review and Moderation (ARM) panel.

DRAFT (WEB VERSION ONLY)



Western Australian Certificate of Education, Sample External Examination
Question/Answer Booklet

**FOOD SCIENCE AND
TECHNOLOGY**
STAGE 2 UNITS 2A–2B

Please place your student identification label in this box

Student Number: In figures

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In words

Time allowed for this paper

Reading time before commencing work: Ten minutes

Working time for paper: Three hours

Material required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet

To be provided by the candidate

Standard items: Pens, pencils, eraser or correction fluid, ruler, highlighter

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Suggested working time	Number of questions available	Number of questions to be attempted	% of paper	Marks
One	20 minutes	20	20	20%	20
Two	100 minutes	10	10	50%	75
Three	30 minutes	1	1	15%	20
Four	30 minutes	6	1	15%	20
Total marks					135

Instructions to candidates

- The rules for the conduct of Curriculum Council examinations are detailed in the *Student Information Handbook*. Sitting this examination implies that you agree to abide by these rules.
- Answer the questions according to the following instructions.
Section A Answer **ALL** questions in the spaces provided in this Question/Answer Booklet.
Section B Answer **ALL** questions in the spaces provided in this Question/Answer Booklet.
Section C Answer **ONE** questions in the pages provided in this Question/Answer Booklet.
Section D Answer **ONE** question from the context you have studied, in the pages provided in this Question/Answer Booklet.
- A blue or black ball point or ink pen should be used.

SECTION ONE—MULTIPLE CHOICE

A multiple choice answer sheet is provided for you to answer questions in this section. Use a **2B, B or HB** pencil for all entries. For each question, shade the box which indicates your answer.

This section has **TWENTY** questions. Attempt **ALL** questions.

Allow approximately 20 minutes for this section. [20 marks].

1.

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<http://www.vcaa.vic.edu.au/vet/programs/hospitality/pastexams/2006/2006vethosp-cc-w.pdf>
(p.2, Question 4)

[From: Victorian Curriculum and Assessment Authority, 2006]

2.

For copyright reasons this question cannot be reproduced in the online version of this document but may be viewed at
http://www.boardofstudies.nsw.edu.au/hsc_exams/hsc2006exams/pdf_doc/food_tech_06.pdf
(p.3, Question 5)

[From: Board of Studies New South Wales, 2006]

3.

For copyright reasons this question cannot be reproduced in the online version of this document but may be viewed at
http://www.boardofstudies.nsw.edu.au/hsc_exams/hsc2006exams/pdf_doc/food_tech_06.pdf
(p.4, Question 8)

[From: Board of Studies New South Wales, 2006]

4. An effect of iron deficiency in the body is

- a) brittle bones.
- b) anaemia.
- c) hyperactivity.
- d) bloating.

5. The importance of including adequate folic acid in the diet is to

- a) prevent depression and irritability.
- b) prevent retarded growth.
- c) assist in growth and nerve development.
- d) assist the formation of red and white blood cells.

6. When cooking cereal products

- a) cellulose softens and absorbs vitamins.
- b) B group vitamins are retained.
- c) starch grains burst and gelatinise.
- d) liquid absorbs the calcium and iron.

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7. Nutrient additives are used in food manufacturing to

- a) replace nutrients lost in food processing.
- b) assist in the absorption of iron.
- c) decrease the shelf life of food products.
- d) prevent skin inflammation and growth retardation.

8. Calcium assists the body to

- a) absorb large amounts of protein in the diet.
- b) participate in aerobic exercise.
- c) clot blood.
- d) increase levels of oestrogen.

9. A nutritional reason for including vegetables in the daily menu plan is

- a) to replace more expensive food items like meat in the diet.
- b) large varieties are available locally and from overseas.
- c) they can be prepared by steaming, pressure cooking, micro-waved and stir frying.
- d) they are low in saturated fats.

10. Two dietary problems associated with excess refined carbohydrate intake are

- a) little heat and energy in the body.
- b) cellulose absorption and flavour.
- c) dental caries and bowel problems.
- d) hunger and excess tiredness.

11. Food manufacturers use additives to enhance the sensory characteristics of their product range because

- a) increased shelf life is important to the consumer.
- b) consumers associate certain colours with flavours.
- c) additives increase the consumer acceptance of the product.
- d) it does not replace the colour that is lost during processing.

Questions 12 and 13 relate to the following information

Consider the ingredients below for a whisked sponge:

3 eggs

75gm caster sugar

75gm plain flour

12. The function of the egg in this recipe is to

- a) add flavour and act as a preservative.
- b) dissolve into a syrup and soften the gluten in the flour.
- c) combine with sugar and form the structure.
- d) provide structure, texture and colour.

13. The change sugar undergoes during the manufacture of a whisked sponge is

- a) gluten coagulates to form structure.
- b) air is trapped when whisked.
- c) coagulates on heating forming structure.
- d) caramelises to add flavour.

14. Freezing is a popular method of food preservation for the consumer today because

- a) freezing maintains the nutritional value of all food products.
- b) frozen food products can be defrosted and re-frozen without damaging quality.
- c) during power failure, freezers not opened will not deteriorate food products.
- d) product sensory qualities are quite well preserved.

15. The Maillard Reaction is

- a) the browning reaction that occurs during roasting, grilling and baking of many food products.
- b) the connective tissue proteins (collagen, elastin and reticulin) and their conversion to gelatine.
- c) the action of the preservatives on the muscle fibres to help keep the product safe for longer.
- d) the reaction of the anti-oxidants used to prevent the fat in the food combining with oxygen and becoming rancid.

16. What does HACCP stand for?

- a) Hazard Analysis Customer Control Points
- b) Hazard Analysis Critical Control Points
- c) Hazard Analysis Consumer Confidence Points
- d) Hazard Analysis Consumer Control Points

17. The main purpose of the HACCP System is to

- a) ensure all food premises have a food safety program approved by the local council.
- b) prevent food spoilage and food poisoning and ensure a safe food supply for consumers.
- c) develop a series of guidelines for most food premises that manufacture, package, transport and sell food.
- d) prevent only the cross contamination of food in the hospitality industry.

18. Many people in society are sensitive to particular foods. The two most common food sensitivities consumers are more prone to today are

- a) lactose intolerance and heart disease.
- b) Coeliac disease and nut allergies.
- c) lactose intolerance and Coeliac disease.
- d) nut allergies and lactose intolerance.

19. Which of the following is NOT a precaution against food contamination?

- a) food should be refrigerated below 5°C.
- b) all chicken should be chopped on the same board.
- c) raw and cooked chicken should not be stored together.
- d) hot food should be kept at 60° or hotter.

20. As many fruits and vegetables age they give off carbon dioxide. The type of packaging used needs to be given consideration to prevent deterioration.

To assist this, modern types of packaging are made from

- a) cardboard to allow the product to breathe.
- b) semi permeable plastic.
- c) hard plastic with holes.
- d) barrier specific packaging.

END OF SECTION ONE

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SECTION TWO—SHORT RESPONSE

This section has **10** questions. Attempt **ALL** questions.

Allow approximately 100 minutes [75 marks]

Question 1**[8 marks]**

- a) Name the electrical appliance shown below.

[1 mark]

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[Diagram from: Heath, McKenzie & Tully, 2005]

Name _____

- b) Give two uses for this electrical appliance when preparing food and in each case explain the advantages of using it, compared to manual handling.

[4 marks]

	Use	Advantage
1		
2		

- c) List three (3) safety precautions that should be taken when using this electrical appliance.

[3 marks]

Question 2**[7 marks]**

Choose a vegetable. Complete the following table showing one dry and one wet cooking method that can be used for preparing the chosen vegetable and explain how this cooking method affects nutrient retention and the change in physical properties of the vegetable.

The vegetable I have chosen is:	
---------------------------------	--

[1 mark]

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(p.7)

[Adapted from: Victorian Curriculum and Assessment Authority, 2006]

[6 marks]**Question 3****[12 marks]**

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<http://www.vcaa.vic.edu.au/vet/programs/hospitality/pastexams/2006/2006vethosp-cc-w.pdf>

(p.10)

For copyright reasons this question cannot be reproduced in the online version of this document but may be viewed at <http://www.vcaa.vic.edu.au/vet/programs/hospitality/pastexams/2006/2006vethosp-cc-w.pdf> (p.10)

[Adapted from: Victorian Curriculum and Assessment Authority, 2006]

Question 4**[6 marks]**

Explain what FSANZ stands for and give one reason why is it important in the food industry.

Question 5**[10 marks]**

- a) Identify and discuss at least two (2) factors that have led to an increase in the number of convenience foods available to the consumer. **[4 marks]**

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- b) List and evaluate three (3) effects that the increased availability of convenience foods may have on the health of individuals. [6 marks]

Question 6**[8 marks]**

Read the following passage and answer the questions.

The caterer takes her cutting board from the boot of her car into the kitchen and cuts chicken and then vegetables for the kebabs. She places the marinated chicken on a plate, uncovered, at the top of the fridge, as she needs to turn it in the marinade every half hour. She places the vegetables and fruit in a deep open plastic bowl on ice in the kitchen sink, as the fridge is full and does not close properly.

- a) Identify and describe at least three potential food safety hazards in the above passage. Explain how each hazard can affect the safety of food. [6 marks]

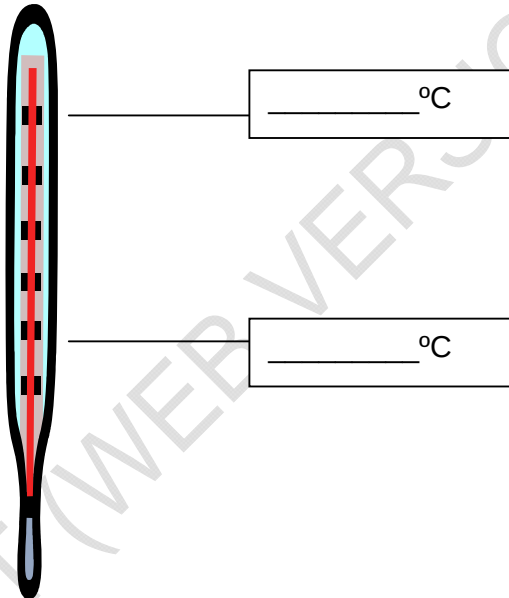
- b) Explain the meaning of cross contamination. Give one example of how this could occur in the kitchen. [2 marks]

Question 7

[6 marks]

Temperature Danger Zone for food.

- a) Fill in the temperatures on the diagram below to show the temperature danger zone. [2 marks]



- b) Explain the safe method of thawing frozen meat. Why is this important and what could happen if meat was thawed incorrectly? [3 marks]

- c) Why should meat be cooked soon after thawing? [1 marks]

Question 8**[6 marks]**

[Adapted from: Victorian Curriculum and Assessment Authority, 2006]

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Question 9**[6 marks]**

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- b) Give one example of an additive and identify the food it is used in. Give reasons for its addition.

[3 marks]

Question 10**[6 marks]**

- a) Explain the difference between food spoilage and food poisoning.

[4 marks]

- b) Identify one of the main causes of food spoilage and explain how this occurs. [2 marks]

END OF SECTION TWO

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END OF SECTION THREE

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SECTION FOUR—CONTEXT SPECIFIC EXTENDED RESPONSE

This section is divided into **THREE** contexts:

Hospitality

OR

Product development

OR

Nutrition and health promotion.

Complete **ONE CONTEXT ONLY**. There are **TWO** questions per context. Answer **ONE** question only.

Allow approximately 30 minutes [20 marks]

HOSPITALITY**Question 1****[20 marks]****Special function to demonstrate hospitality skills**

Design an innovative menu suitable for a function to be prepared and served by your class. The meal should include three (3) courses and be served with hot or cold drinks. It should be economical to produce (no prawns or crayfish!)

Select one of the following themes for your function:

- Youth Week to celebrate youth achievements
- Promotion of local produce
- Melbourne Cup luncheon
- Farewell meal for visiting overseas or regional students
- Thank you to local businesses and community people to promote the school.

Include the following in your response

- identify the target market and give numbers
- type of meal to be served (buffet or sit down)
- explanation of the functional, aesthetic, social and environmental considerations that have influenced your decisions

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HOSPITALITY

Question 2

[20 marks]

[Adapted from: Perraton et al., 2005]

“Meals in a Moment”

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PRODUCT DEVELOPMENT

Question 1

[20 marks]

Healthy Meal Deals

Design a 'healthy meal deal' to be sold in a local primary school canteen for lunch. This is to be part of a 'Healthy Eating Campaign'. They are to be economical to produce and cost no more than \$5.00 each.

- a) Discuss important factors to consider when designing the meal deal.
- b) List foods and drink to be included in the 'meal deal' and justify the choices you have made.
- c) Consider functional, aesthetic, social, environmental, ethical, political and health issues that have influenced your decisions.

Handwriting practice lines for the answer.

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PRODUCT DEVELOPMENT

Question 2

[20 marks]

Design a “healthy – fresh is best” food wrap product to be sold at the local high school canteen using a variety of food products. The products are to be low costing.

- Identify your target market.
- Draw and label the wrap to address the design criteria.
- Identify and discuss the food presentation and packaging of this product for the target market.
- Justify your choice by considering the target market and the functional, aesthetic, social and environmental issues that have influenced your decision.
- Describe ways to evaluate the success of the product and identify product improvements.

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NUTRITION AND HEALTH PROMOTION

Question 1

[20 marks]

Promoting Fruit and Vegetables for Children

Design an innovative plan for a health campaign to promote the inclusion of more fresh **fruit and vegetables** in children's diets. The campaign is to be aimed at Western Australian schools and produced at minimum cost.

- In your plan, consider functional, aesthetic, social, environmental, ethical and political issues that have influenced your decisions.
- Identify strategies for measuring the success of the campaign.

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NUTRITION AND HEALTH PROMOTION

Question 2

[20 marks]

The media, busy life style, income and the availability of convenience and fast foods have encouraged poor diets, leading to increased levels of obesity in Australia amongst children and adults.

Evaluate this statement in relation to the societal issues and trends that impact on the health and wellbeing of individuals and communities.

Discuss the impact of the issues and trends on the supply of state and federal resources and services.

In your response consider social, functional, health, nutritional, environmental and economical issues.

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END OF PAPER

ACKNOWLEDGEMENTS

SECTION ONE

- Question 1** Victorian Curriculum and Assessment Authority. (2006). *VCE VET Hospitality (Operations) Commercial Cookery: Written examination: Victorian Certificate of Education 2006* (p. 2). Retrieved October, 2007, from <http://www.vcaa.vic.edu.au/vet/programs/hospitality/pastexams/2006/2006vethosp-cc-w.pdf>.
- Question 2** Board of Studies New South Wales. (2006). *Food Technology: 2006 Higher School Certificate Examination* (p. 3). Retrieved October, 2007, from http://www.boardofstudies.nsw.edu.au/hsc_exams/hsc2006exams/pdf_doc/food_tech_06.pdf.
- Question 3** Board of Studies New South Wales. (2006). *Food Technology: 2006 Higher School Certificate Examination* (p. 4). Retrieved October, 2007, from http://www.boardofstudies.nsw.edu.au/hsc_exams/hsc2006exams/pdf_doc/food_tech_06.pdf.

SECTION TWO

- Question 1** Diagram from: Heath, G., McKenzie, H., & Tully, L. (2005). *Food solutions: Food & technology units 1 & 2*. Sydney: Pearson Longman, p. 27.
- Question 2** Adapted from: Victorian Curriculum and Assessment Authority. (2006). *Food and Technology: Written examination: Victorian Certificate of Education 2006* (p. 7). Retrieved October, 2007, from <http://www.vcaa.vic.edu.au/vce/studies/foodtech/pastexams/2006food-tech-w.pdf>.
- Question 3** Adapted from: Victorian Curriculum and Assessment Authority. (2006). *VCE VET Hospitality (Operations) Commercial Cookery: Written examination: Victorian Certificate of Education 2006* (p. 10). Retrieved October, 2007, from <http://www.vcaa.vic.edu.au/vet/programs/hospitality/pastexams/2006/2006vethosp-cc-w.pdf>.
- Question 8** Adapted from: Victorian Curriculum and Assessment Authority. (2006). *Food and Technology: Written examination: Victorian Certificate of Education 2006* (p. 6). Retrieved October, 2007, from <http://www.vcaa.vic.edu.au/vce/studies/foodtech/pastexams/2006food-tech-w.pdf>.
- Question 9** Adapted from: State Examinations Commission. (2002). *Home Economics (Scientific and Social) Higher Level: An Roinn Oideachais Agus Eolaiochta: Leaving Certificate Examination, 2002* (p. 2). Retrieved October, 2007, from <http://www.examinations.ie/archive/exampapers/2002/LC030ALPO00EV.pdf>.

SECTION FOUR Hospitality

- Question 2** Adapted from: Perraton, G., et al. (2005). *Food and technology 1*. Milton, Qld: John Wiley & Sons, p. 191.